Experiment No:3 **Date:**

Water Level Detection and Alert System

Aim: Write a program to interface water level detection and send alert

Components Required:

- Water level sensor
- Arduino Uno
- Connecting Wires
- LEDs
- Resistors
- Breadboard

Theory:

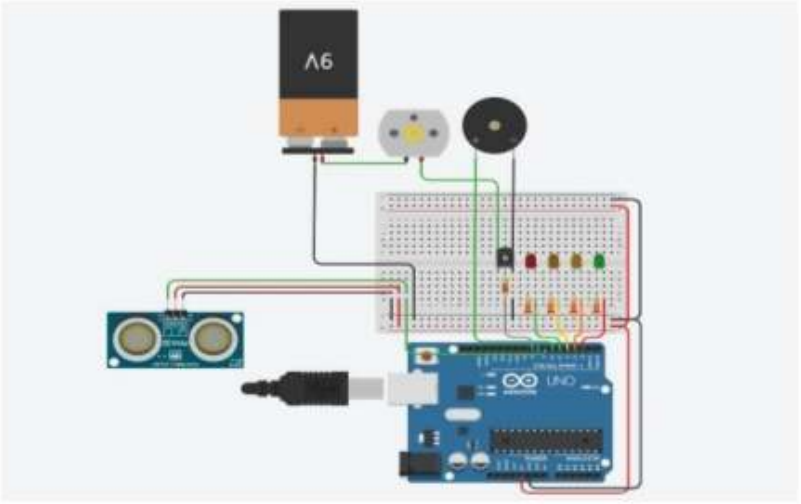
Water level controller using Arduino. The circuit displays the level of water in the tank and switches the motor ON when the water level goes below a predetermined level. The circuit automatically switches the motor OFF when the tank is full. A beep sound is generated when the level in the tank is full.

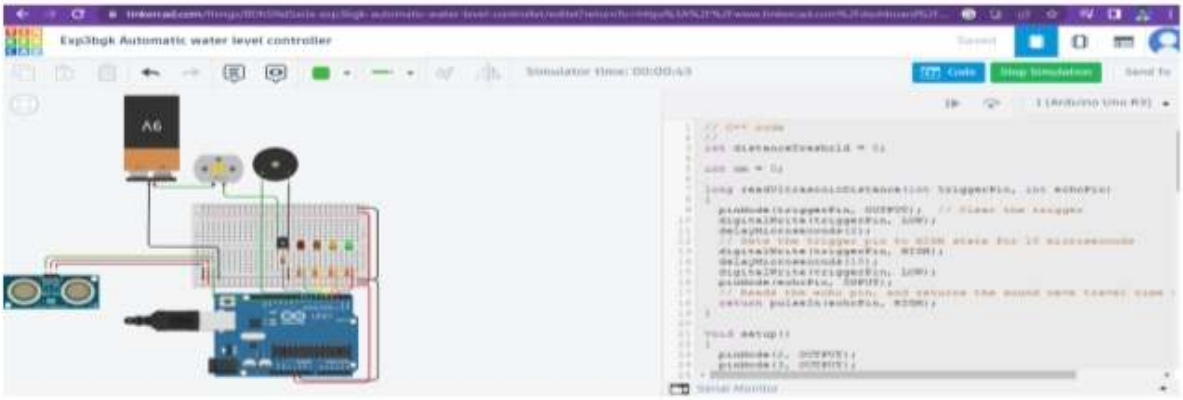
Circuit diagram I have used 4 LEDs for 4 different levels.

1. Green led is connected with pin 2 indicating the level 1 and pump will start automatically at level 1
2. Orange led is connected with pin 3 indicating the level 2
3. Yellow led is connected with pin 4 indicating the level 3
4. Red led is connected with pin 5 indicating the level 4 and buzzer is also connected along with level 4 which means the tank is full.

At level 4 all the 4 less will glow , the buzzer will also beep and the pump will stop automatically.

Circuit Diagram:






```
//Program for  Water Level Detection and Alert System
//int distanceTreshold = 0;
int cm = 0;
long readUltrasonicDistance(int triggerPin, int echoPin)
{
  pinMode(triggerPin, OUTPUT);  // Clear the trigger
  digitalWrite(triggerPin, LOW);
  delayMicroseconds(2);
  // Sets the trigger pin to HIGH state for 10 microseconds
  digitalWrite(triggerPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(triggerPin, LOW);
  pinMode(echoPin, INPUT);
  // Reads the echo pin, and returns the sound wave travel time in microseconds
  return pulseIn(echoPin, HIGH);
}
void setup()
{
  pinMode(2, OUTPUT);
  pinMode(3, OUTPUT);
  pinMode(4, OUTPUT);
  pinMode(5, OUTPUT);
  pinMode(8, OUTPUT);
  pinMode(6, OUTPUT);
}
void loop()
{
  distanceTreshold = 300;
  cm = 0.01723 * readUltrasonicDistance(7, 7);
  if (cm > distanceTreshold) {
    digitalWrite(2, LOW);
    digitalWrite(3, LOW);
    digitalWrite(4, LOW);
    digitalWrite(5, LOW);
    digitalWrite(8, HIGH);
    noTone(6);
  }
  if (cm <= distanceTreshold && cm > distanceTreshold - 75) {
    digitalWrite(2, LOW);
    digitalWrite(3, LOW);
    digitalWrite(4, LOW);

    digitalWrite(5, LOW);
    digitalWrite(8, HIGH);
```

```
  noTone(6);
  delay(1000); // Wait for 1000 millisecond(s)
  digitalWrite(2, HIGH);
 }
 if (cm <= distanceTreshold - 75 && cm > distanceTreshold - 150) {
  digitalWrite(2, HIGH);
  digitalWrite(3, LOW);
  digitalWrite(4, LOW);
  digitalWrite(5, LOW);
  digitalWrite(8, HIGH);
   noTone(6);
  delay(1000); // Wait for 1000 millisecond(s)
  digitalWrite(3, HIGH);
 }
 if (cm <= distanceTreshold - 150 && cm > distanceTreshold - 225) {
  digitalWrite(2, HIGH);
  digitalWrite(3, HIGH);
  digitalWrite(4, LOW);
  digitalWrite(5, LOW);
  digitalWrite(8, HIGH);
  noTone(6);
  delay(1000); // Wait for 1000 millisecond(s)
  digitalWrite(4, HIGH);
 }
 if (cm <= distanceTreshold - 225 && cm > distanceTreshold - 285) {
  digitalWrite(2, HIGH);
  digitalWrite(3, HIGH);
  digitalWrite(4, HIGH);
  digitalWrite(5, LOW);
  digitalWrite(8, HIGH);
  delay(1000); // Wait for 1000 millisecond(s)
  digitalWrite(5, HIGH);
 }
 if (cm <= distanceTreshold - 285) {
  digitalWrite(2, HIGH);
  digitalWrite(3, HIGH);
  digitalWrite(4, HIGH);
  digitalWrite(5, HIGH);
  digitalWrite(8, LOW);
  tone(6, 523, 1000); // play tone 60 (C5 = 523 Hz)
 }}
```

Procedure:

1. Open tinker cad
2. Setup the circuit as per circuit diagram
3. Connect the components as shown in the connection diagram.
4. Connect power supply
5. Verify the output varying voltage and current.

Result:

References:

https://www.instructables.com/Water-Level-Indicator-Using-Arduino-Tinker-Ca